TrustCheck: AI models

Every model the prototype runs, what it does, its architecture, parameters and layers.

Parameter and layer counts were **counted from the model weights on the team laptop** (RTX 4050, 6 GB), except Whisper large-v3, which uses its published figures. In total about **2.9 billion parameters** run locally, together in about 5.5 GB of GPU memory: no cloud AI service is needed. A voice note is checked in about 2.5 s, a video in about 4.5 s.

1. The models at a glance

Job	Model	Architecture	Parameters	Layers
Cloned voice	pellav2 (Sadanie)	w2v-BERT 2.0 Conformer encoder + learned mix of 25 hidden states + linear head 1024 → 1	580.5 M	24 Conformer, width 1024, 16 heads
Voiceprint	ECAPA-TDNN (SpeechBrain, VoxCeleb)	TDNN → SE-Res2Net blocks → attentive stats pooling → 192-d embedding	22.2 M	3 SE-Res2Net blocks + in/out TDNN
Transcript	Whisper large-v3 (faster-whisper, int8)	Transformer encoder-decoder, 128 mel bins	1,550 M (published)	32 encoder + 32 decoder, width 1280, 20 heads
Face swap (video)	GenD (WACV 2026)	CLIP ViT-L/14 vision encoder + linear probe (2 classes), team calibration	304.0 M	24 ViT, width 1024, 16 heads, patch 14
AI image	UnivFD (CVPR 2023)	CLIP ViT-L/14 (frozen) + linear head 768 → 1, retrained by the team	304.0 M (head 769)	24 ViT + 1 linear
Face detection	InsightFace SCRFD-10G (buffalo_l)	Single-stage CNN face detector	4.2 M	58 conv
Face match	InsightFace ArcFace (WebFace600K)	ResNet-50 → 512-d embedding	43.6 M	53 conv
Scam meaning	paraphrase-multilingual -MiniLM-L12-v2 + team logistic regression	Small BERT-style Transformer → 384-d sentence vector → linear 1 × 384	117.7 M	12 Transformer, width 384
Scam wording	Team-trained TF-IDF classifier	Char 2-5 and word 1-2 n-grams → calibrated logistic regression	62,273 features	1 linear
Screenshot text	RapidOCR (PP-OCRv4 mobile)	Text detection + angle classifier + recognition CNNs	small (4.6 + 0.6 + 10.6 MB files)	62 + 53 + 38 conv

2. Each model in detail

pellav2: is the voice cloned?

Backbone: Meta's **w2v-BERT 2.0**, a 24-layer Conformer speech encoder (width 1024, 16 attention heads) pretrained on 4.5 M hours of speech. pellav2 learns a weighted mix of all 25 hidden states and a linear head (1024 → 1). Input: 16 kHz mono, loudness-normalised to -23 LUFS, 20 Hz high-pass, 4-second windows. TrustCheck scores every window and takes the mean of the worst third, so a short cloned segment inside a real call still counts. **Measured:** 120/120 F5-TTS clones caught (incl. WhatsApp Opus), 0.8% of real clips flagged.

ECAPA-TDNN: is it the same person's voice?

Time-delay network with 3 squeeze-and-excitation Res2Net blocks, multi-layer feature aggregation and attentive statistics pooling, giving a 192-dimension speaker embedding. Compared by cosine similarity to the enrolled voiceprint (threshold 0.40). **Limit:** clones of a speaker were accepted 100% of the time, so the voiceprint never decides alone.

Whisper large-v3: what was said?

OpenAI's encoder-decoder Transformer (32 + 32 layers, width 1280, 20 heads, 128 mel bins), run through faster-whisper with int8 weights and voice-activity filtering, with an initial prompt tuned for Indian money talk. The transcript goes through every text check and, for the video YES, must contain the live code word.

GenD: is the face swapped?

CLIP ViT-L/14 image encoder (24 layers, width 1024, 16 heads, 14 px patches, 224 px input) with a linear probe to real / fake. InsightFace finds and crops the face in 32 sampled frames; the clip score is the mean of the worst third; a Platt calibration fitted by the team on 240 separate videos maps it to a probability. **Measured:** 91% of swaps caught at 3% real flagged (re-encoded: 88% / 35%).

UnivFD: is the image AI-generated?

The same CLIP ViT-L/14 backbone, frozen, with a linear head on the 768-d image embedding. The team refit the head on 1,000 COCO photos vs 1,000 SD-Turbo images: AI images caught went from 11% to 97%, false alarms from 3% to 6%. Combined with C2PA and metadata checks.

InsightFace buffalo_l: where is the face, and whose is it?

SCRFD-10G (4.2 M parameters, 58 conv layers) detects faces; ArcFace on ResNet-50 (43.6 M, trained on WebFace600K) makes a 512-d face embedding, compared by cosine similarity (threshold 0.40) to the enrolled photo. Runs on the GPU through ONNX Runtime.

Message models: is the text a scam?

Two classifiers run side by side. A **TF-IDF** model trained by the team (62,273 character and word n-gram features, calibrated logistic regression) on 7,059 messages incl. 41 scam and 43 genuine Indian templates: AUROC 0.995 on SMS spam, 0.906 on unseen Indian templates. A **semantic** model: multilingual MiniLM-L12 (12 layers, 384-d, 117.7 M) with a team logistic regression; averaged with TF-IDF it reaches AUROC 0.924 on hand-written held-out messages. Plus 24 scam patterns, link intelligence and a Hindi / Hinglish lexicon.

RapidOCR: what does the screenshot say?

PaddleOCR v4 mobile models (detection, angle classification, recognition) in ONNX. A screenshot with 40+ characters of text is read and sent through the text checks instead of image forensics.

3. How the models' scores become one answer

Each detector returns a score from 0 to 1 and a reason. Fusion adds them in log-odds from a 5% prior, weighs them, and multiplies by how much damage the request could do. Policy rules then set a floor no detector can talk its way under.

$R = 100 \times \sigma(b + \Sigma\, w_i \cdot \ln(s_i / (1 - s_i))) \times \text{Impact}$, $b = \ln(0.05 / 0.95) = -2.944$

Signal	Weight	Signal	Weight
Known mule account	1.5	Face swap (GenD)	0.8
Bad URL	1.2	Voice mismatch (ECAPA)	0.8
Cloned voice (pellav2)	1.0	Scam pattern	0.8
Context vs Trust Graph	1.0	AI image (UnivFD)	0.7
Link risk	1.0	Face mismatch (ArcFace)	0.6
Lip-sync	0.5	Text classifier	0.4

Impact of the ask: big payment, crypto or gift cards 1.0 · install an app 0.9 · OTP or password 0.8 · click a link 0.7 · personal data 0.6 · call back 0.5 · share documents 0.4 · nothing asked 0.2. **Floors:** an unregistered channel, new account, mule, bad link or digital-arrest pattern is never below HIGH (60), rising to 75 at ₹1 lakh and 90 from ₹10 lakh. **Bands:** LOW < 30, MEDIUM 30–59, HIGH ≥ 60.

Worked examples: a fake CFO asking for a wire scores 99.9; the same fake sending a birthday wish scores 9.8; a perfect fake that fools every detector scores 17.3 by the maths and is lifted to HIGH (60) by policy.

4. Video YES: models reused to prove a YES is real

For a request above the approval limit, one tap of YES can't release it (a hacked phone could tap it). The person records a 7-second selfie video saying a code word that changes every 30 seconds. **Required:** GenD finds no face swap, ArcFace matches the enrolled photo, Whisper heard the live code word. **Supporting, at least one of:** pellav2 real voice, ECAPA voiceprint match, lip-sync. The clip is deleted after scoring.

5. Measured results and limits

Model / check	Test set	Result	Limit
pellav2	120 clones of 40 speakers + 360 real clips, MP3 and Opus	AUROC 1.000, 120/120 caught, 0.8% real flagged	One cloning engine; phone-browser audio scored a real voice 0.96–0.98 synthetic
ECAPA voiceprint	Same clones vs the enrolled voice	Clones accepted 100%	Never decides alone
GenD (calibrated)	300 FaceForensics++-style videos	91% caught, 3% real flagged	Re-encoded 88% / 35%; live swaps untested
UnivFD (retrained head)	600 held-out COCO vs SD-Turbo	97% caught, 6% false alarms	One generator family
Message pipeline	4 hand-written sets never seen in training	58/80 scams HIGH, 60/60 genuine LOW	Small sets written by the team
Robustness	44 hostile inputs (Unicode tricks, corrupt files)	44/44 handled	Not load testing

Privacy: raw uploads and video-YES clips are deleted after analysis; only voice (192-d) and face (512-d) embeddings are stored. All models run on the laptop; nothing is sent to a cloud AI service.